Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2003-II-Q03] | Part A (Q1-Q5)

Count seven-letter words over $\{A, B, C\}$ that contain none of the adjacent blocks AB, BC, CA.

用 A, B, C 组成七字母串，禁止相邻块 AB, BC, CA，共有多少个?

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Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2004-II-Q04] | Part A (Q1-Q5)

How many positive decimal integers below 10000 use at most two distinct digit values?

小于一万的正十进制整数中，有多少个最多使用两种不同数位?

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Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2005-I-Q05] | Part A (Q1-Q5)

Stack four identical gold and four identical silver coins. Each has one engraved face, independently orientable up or down. Count distinguishable stacks with no adjacent engraved faces facing one another.

将四枚相同金币与四枚相同银币叠成一柱，每枚仅一面有雕刻，可朝上或朝下。求相邻两枚雕刻面不相对的不同叠法数。

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受限排列

[AIME-2006-II-Q04] | Part A (Q1-Q5)

Count permutations of $1, \ldots, 12$ satisfying $a_1 > a_2 > \cdots > a_6 < a_7 < \cdots < a_{12}$.

求一至十二的排列中满足 $a_1 > a_2 > \cdots > a_6 < a_7 < \cdots < a_{12}$ 的个数。

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Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2007-II-Q01] | Part A (Q1-Q5)

Form five-character strings from the multiset $\{A, I, M, E, 2, 0, 0, 7\}$, never exceeding the supplied multiplicities. If there are N distinct strings, find $N/10$.

从多重集合 $\{A, I, M, E, 2, 0, 0, 7\}$ 中取字符排成长五的串，各字符不超过原有次数。若不同串数为 N，求 $N/10$。

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受限排列

[AIME-2011-I-Q05] | Part A (Q1-Q5)

Label a regular nonagon with digits 1 through 9, each once, so every three consecutive labels have sum divisible by 3. Count arrangements up to rotation, with reflections distinct.

将数字一至九各用一次标在正九边形顶点，使任意连续三顶点之和可被三整除。旋转视为相同，反射仍区分，求排列数。

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受限排列

[AIME-2012-I-Q03] | Part A (Q1-Q5)

Nine distinct diners order three meals of each of three types. Count the assignments of these meal types to diners for which exactly one diner gets the requested type.

九名不同食客各点三种餐之一，每种恰有三人。求分配三类餐品的方法数，使恰一人收到自己所点的餐。

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受限排列

[AIME-2019-I-Q04] | Part A (Q1-Q5)

A team starts eleven fixed players from twenty-two. It may make zero through three sequential substitutions: any current player leaves, an unused substitute enters, and removed players never return. An entering substitute may later leave. Count substitution sequences modulo 1000.

二十二人队先固定十一人上场，可依次换零至三次，每次任选当前一人离场、未上过场的替补一人进入，离场者不得返回，进入者之后也可被换下。求换人序列数除以一千的余数。

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受限排列

[AIME-2019-II-Q05] | Part A (Q1-Q5)

Four distinguishable ambassadors and their four distinct paired advisors sit in twelve numbered circular chairs. Ambassadors use even chairs, each advisor sits adjacent to their ambassador, and no chair is shared. Find the seating count modulo 1000.

四名不同大使及各自不同顾问坐十二个编号圆桌座位，大使只能坐偶数位，顾问须邻其大使，不得重座。求安排数除以一千的余数。

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受限排列

[AIME-2020-I-Q05] | Part A (Q1-Q5)

Permute the six distinct labels $1, 2, 3, 4, 5, 6$. How many permutations permit deletion of one label so that the remaining sequence is strictly increasing or strictly decreasing?

排列 $1, 2, 3, 4, 5, 6$ 六个不同标号。若删去一个标号后，其余五个数可严格递增或严格递减，求这样的排列数。

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受限排列

[AIME-2021-II-Q03] | Part A (Q1-Q5)

For a permutation $(x_1, \ldots, x_5)$ of $1, 2, 3, 4, 5$, form $\sum_{i=1}^{5} x_i x_{i+1} x_{i+2}$ with cyclic indices. How many permutations make this sum divisible by 3?

将 $1, 2, 3, 4, 5$ 排成 $(x_1, \ldots, x_5)$，下标循环使用。求使 $\sum_{i=1}^{5} x_i x_{i+1} x_{i+2}$ 被 3 整除的排列数。

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受限排列

[AIME-2025-I-Q03] | Part A (Q1-Q5)

Nine distinct people each choose one of three flavors. All flavors must occur, with first-flavor count greater than second-flavor count greater than third-flavor count. Find the number of assignments modulo 1000.

九个不同的人各选三种口味之一，每种都至少有人选择，且第一种人数大于第二种，第二种大于第三种。求合法选择数除以 1000 的余数。

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Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2004-I-Q06] | Part B (Q6-Q10)

Count four-digit integers with distinct digits satisfying $a_1 < a_2 > a_3 < a_4$.

求四个数位互异且满足 $a_1 < a_2 > a_3 < a_4$ 的四位整数个数。

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受限排列

[AIME-2023-II-Q07] | Part B (Q6-Q10)

Color each vertex of a regular twelve-gon red or blue. Among the 2^{12} labeled colorings, how many contain no rectangle whose four vertices share one color?

将正十二边形每个顶点涂成红或蓝。2^{12} 种按位置区分的染色中，不存在四顶点同色矩形的共有多少种?

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受限排列

[AIME-2025-I-Q10] | Part B (Q6-Q10)

Fill a 3×9 grid with digits $1, \ldots, 9$ so each row and each of the three consecutive 3×3 blocks contains every digit once. If the number of fillings is $\prod p_j^{e_j}$, find $\sum p_j e_j$.

用数字 1 至 9 填满 3×9 方格，每行及依次排列的三个 3×3 宫都恰含每个数字一次。将填法数分解为 $\prod p_j^{e_j}$，求 $\sum p_j e_j$。

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受限排列

[AIME-2026-I-Q07] | Part B (Q6-Q10)

Count onto maps $\pi : \{1, \ldots, 6\} \to \{1, \ldots, 6\}$ satisfying $\pi^6(a) = a$ for all a.

求满射 $\pi : \{1, \ldots, 6\} \to \{1, \ldots, 6\}$ 的个数，使每个 a 均满足六次迭代 $\pi^6(a) = a$。

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